



SITOP PSU100S/1AC/24VDC/10A

SITOP PSU100S 24 V/10 A stabilized power supply input: 120/230 V AC output: 24 V DC/10 A

| input                                                                                      |                                                                   |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| type of the power supply network                                                           | 1-phase AC                                                        |
| supply voltage at AC                                                                       | Automatic range selection                                         |
| supply voltage                                                                             | 120 V/230 V                                                       |
| input voltage 1 at AC                                                                      | 85 ... 132 V                                                      |
| input voltage 2 at AC                                                                      | 170 ... 264 V                                                     |
| wide range input                                                                           | No                                                                |
| overvoltage overload capability                                                            | $2.3 \times V_{in}$ rated, 1.3 ms                                 |
| buffering time for rated value of the output current in the event of power failure minimum | 20 ms                                                             |
| operating condition of the mains buffering                                                 | at $V_{in} = 93/187$ V                                            |
| line frequency                                                                             | 50/60 Hz                                                          |
| line frequency                                                                             | 47 ... 63 Hz                                                      |
| input current                                                                              |                                                                   |
| • at rated input voltage 120 V                                                             | 4.49 A                                                            |
| • at rated input voltage 230 V                                                             | 1.91 A                                                            |
| current limitation of inrush current at 25 °C maximum                                      | 60 A                                                              |
| I <sup>2</sup> t value maximum                                                             | 5.6 A <sup>2</sup> s                                              |
| fuse protection type                                                                       | T 6.3 A/250 V (not accessible)                                    |
| fuse protection type in the feeder                                                         | Recommended miniature circuit breaker: from 10 A characteristic C |
| output                                                                                     |                                                                   |
| voltage curve at output                                                                    | Controlled, isolated DC voltage                                   |
| output voltage at DC rated value                                                           | 24 V                                                              |
| output voltage                                                                             |                                                                   |
| • at output 1 at DC rated value                                                            | 24 V                                                              |
| output voltage adjustable                                                                  | Yes; via potentiometer                                            |
| adjustable output voltage                                                                  | 22.8 ... 28 V                                                     |
| relative overall tolerance of the voltage                                                  | 3 %                                                               |
| relative control precision of the output voltage                                           |                                                                   |
| • on slow fluctuation of input voltage                                                     | 0.1 %                                                             |
| • on slow fluctuation of ohm loading                                                       | 1 %                                                               |
| residual ripple                                                                            |                                                                   |
| • maximum                                                                                  | 150 mV                                                            |
| • typical                                                                                  | 20 mV                                                             |
| voltage peak                                                                               |                                                                   |
| • maximum                                                                                  | 240 mV                                                            |
| • typical                                                                                  | 160 mV                                                            |
| display version for normal operation                                                       | Green LED for 24 V OK                                             |
| type of signal at output                                                                   | Relay contact (NO contact, rating 60 V DC/ 0.3 A) for "24 V OK"   |
| behavior of the output voltage when switching on                                           | Overshoot of $V_{out} < 3$ %                                      |

|                                                                                                                 |                                                                                   |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| response delay maximum                                                                                          | 0.3 s                                                                             |
| voltage increase time of the output voltage                                                                     |                                                                                   |
| • typical                                                                                                       | 20 ms                                                                             |
| output current                                                                                                  |                                                                                   |
| • rated value                                                                                                   | 10 A                                                                              |
| • rated range                                                                                                   | 0 ... 12 A; 12 A up to +45°C; +60 ... +70 °C: Derating 3%/K                       |
| supplied active power typical                                                                                   | 288 W                                                                             |
| short-term overload current                                                                                     |                                                                                   |
| • on short-circuiting during the start-up typical                                                               | 32 A                                                                              |
| • at short-circuit during operation typical                                                                     | 32 A                                                                              |
| duration of overloading capability for excess current                                                           |                                                                                   |
| • on short-circuiting during the start-up                                                                       | 1 000 ms                                                                          |
| • at short-circuit during operation                                                                             | 1 000 ms                                                                          |
| bridging of equipment                                                                                           | Yes                                                                               |
| number of parallel-switched equipment resources for increasing the power                                        | 2                                                                                 |
| <b>efficiency</b>                                                                                               |                                                                                   |
| efficiency in percent                                                                                           | 90 %                                                                              |
| power loss [W]                                                                                                  |                                                                                   |
| • at rated output voltage for rated value of the output current typical                                         | 25 W                                                                              |
| <b>closed-loop control</b>                                                                                      |                                                                                   |
| relative control precision of the output voltage with rapid fluctuation of the input voltage by +/- 15% typical | 0.3 %                                                                             |
| relative control precision of the output voltage at load step of resistive load 10/90/10 % typical              | 3 %                                                                               |
| setting time                                                                                                    |                                                                                   |
| • load step 10 to 90% typical                                                                                   | 1 ms                                                                              |
| • load step 90 to 10% typical                                                                                   | 1 ms                                                                              |
| <b>protection and monitoring</b>                                                                                |                                                                                   |
| design of the overvoltage protection                                                                            | protection against overvoltage in case of internal fault $V_{out} < 33 \text{ V}$ |
| property of the output short-circuit proof                                                                      | Yes                                                                               |
| design of short-circuit protection                                                                              | Constant current characteristic                                                   |
| response value current limitation                                                                               | 12 ... 14.6 A                                                                     |
| overcurrent overload capability                                                                                 |                                                                                   |
| • in normal operation                                                                                           | overload capability 150 % $I_{out}$ rated up to 5 s/min                           |
| enduring short circuit current RMS value                                                                        |                                                                                   |
| • typical                                                                                                       | 14.6 A                                                                            |
| <b>safety</b>                                                                                                   |                                                                                   |
| galvanic isolation between input and output                                                                     | Yes                                                                               |
| galvanic isolation                                                                                              | Safety extra-low output voltage $U_{out}$ acc. to EN 60950-1 and EN 50178         |
| operating resource protection class                                                                             | Class I                                                                           |
| leakage current                                                                                                 |                                                                                   |
| • maximum                                                                                                       | 3.5 mA                                                                            |
| • typical                                                                                                       | 0.8 mA                                                                            |
| protection class IP                                                                                             | IP20                                                                              |
| <b>EMC</b>                                                                                                      |                                                                                   |
| standard                                                                                                        |                                                                                   |
| • for emitted interference                                                                                      | EN 55022 Class B                                                                  |
| • for mains harmonics limitation                                                                                | EN 61000-3-2                                                                      |
| • for interference immunity                                                                                     | EN 61000-6-2                                                                      |
| <b>standards, specifications, approvals</b>                                                                     |                                                                                   |
| certificate of suitability                                                                                      |                                                                                   |
| • CE marking                                                                                                    | Yes                                                                               |
| • UL approval                                                                                                   | Yes; cULus-Listed (UL 508, CSA C22.2 No. 107.1), File E197259                     |
| • UKCA marking                                                                                                  | Yes                                                                               |
| • EAC approval                                                                                                  | Yes                                                                               |
| • NEC Class 2                                                                                                   | No                                                                                |
| type of certification                                                                                           |                                                                                   |
| • BIS                                                                                                           | Yes; R-41183539, R-41188271                                                       |
| • CB-certificate                                                                                                | Yes                                                                               |

|                                                                               |                                                                                                   |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| MTBF at 40 °C                                                                 | 1 614 510 h                                                                                       |
| <b>standards, specifications, approvals hazardous environments</b>            |                                                                                                   |
| certificate of suitability                                                    |                                                                                                   |
| • IECEx                                                                       | No                                                                                                |
| • ATEX                                                                        | No                                                                                                |
| • ULhazloc approval                                                           | No                                                                                                |
| • FM registration                                                             | No                                                                                                |
| <b>standards, specifications, approvals marine classification</b>             |                                                                                                   |
| shipbuilding approval                                                         | Yes                                                                                               |
| Marine classification association                                             |                                                                                                   |
| • American Bureau of Shipping Europe Ltd. (ABS)                               | No                                                                                                |
| • French marine classification society (BV)                                   | Yes                                                                                               |
| • Det Norske Veritas (DNV)                                                    | Yes                                                                                               |
| • Lloyds Register of Shipping (LRS)                                           | No                                                                                                |
| <b>standards, specifications, approvals Environmental Product Declaration</b> |                                                                                                   |
| Environmental Product Declaration                                             | Yes                                                                                               |
| global warming potential [CO2 eq]                                             |                                                                                                   |
| • total                                                                       | 699.2 kg                                                                                          |
| • during manufacturing                                                        | 20.6 kg                                                                                           |
| • during operation                                                            | 677.8 kg                                                                                          |
| • after end of life                                                           | 0.6 kg                                                                                            |
| <b>ambient conditions</b>                                                     |                                                                                                   |
| ambient temperature                                                           |                                                                                                   |
| • during operation                                                            | -25 ... +70 °C; with natural convection                                                           |
| • during transport                                                            | -40 ... +85 °C                                                                                    |
| • during storage                                                              | -40 ... +85 °C                                                                                    |
| environmental category according to IEC 60721                                 | Climate class 3K3, 5 ... 95% no condensation                                                      |
| <b>connection method</b>                                                      |                                                                                                   |
| type of electrical connection                                                 | screw terminal                                                                                    |
| • at input                                                                    | L, N, PE: 1 screw terminal each for 0.5 ... 2.5 mm <sup>2</sup> single-core/finely stranded       |
| • at output                                                                   | +, -: 2 screw terminals each for 0.5 ... 2.5 mm <sup>2</sup>                                      |
| • for auxiliary contacts                                                      | Alarm signals: 2 screw terminals for 0.5 ... 2.5 mm <sup>2</sup>                                  |
| • for signaling contact                                                       | 2 screw terminals for 0.5 ... 2.5 mm <sup>2</sup>                                                 |
| <b>mechanical data</b>                                                        |                                                                                                   |
| width × height × depth of the enclosure                                       | 70 × 125 × 120 mm                                                                                 |
| installation width × mounting height                                          | 70 mm × 225 mm                                                                                    |
| required spacing                                                              |                                                                                                   |
| • top                                                                         | 50 mm                                                                                             |
| • bottom                                                                      | 50 mm                                                                                             |
| • left                                                                        | 0 mm                                                                                              |
| • right                                                                       | 0 mm                                                                                              |
| fastening method                                                              | Snaps onto DIN rail EN 60715 35x7.5/15                                                            |
| • DIN-rail mounting                                                           | Yes                                                                                               |
| • S7 rail mounting                                                            | No                                                                                                |
| • wall mounting                                                               | No                                                                                                |
| housing can be lined up                                                       | Yes                                                                                               |
| net weight                                                                    | 0.8 kg                                                                                            |
| <b>accessories</b>                                                            |                                                                                                   |
| electrical accessories                                                        | Buffer module                                                                                     |
| mechanical accessories                                                        | Device identification label 20 mm × 7 mm, pale turquoise 3RT1900-1SB20                            |
| <b>further information internet links</b>                                     |                                                                                                   |
| internet link                                                                 |                                                                                                   |
| • to website: Industry Mall                                                   | <a href="https://mall.industry.siemens.com">https://mall.industry.siemens.com</a>                 |
| • to web page: selection aid TIA Selection Tool                               | <a href="https://www.siemens.com/tstcloud">https://www.siemens.com/tstcloud</a>                   |
| • to web page: power supplies                                                 | <a href="https://siemens.com/sitop">https://siemens.com/sitop</a>                                 |
| • to website: CAx-Download-Manager                                            | <a href="https://siemens.com/cax">https://siemens.com/cax</a>                                     |
| • to website: Industry Online Support                                         | <a href="https://support.industry.siemens.com">https://support.industry.siemens.com</a>           |
| <b>additional information</b>                                                 |                                                                                                   |
| other information                                                             | Specifications at rated input voltage and ambient temperature +25 °C (unless otherwise specified) |

security information

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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Classifications

|        | Version | Classification |
|--------|---------|----------------|
| eClass | 14      | 27-04-07-01    |
| eClass | 12      | 27-04-07-01    |
| eClass | 9.1     | 27-04-07-01    |
| eClass | 9       | 27-04-07-01    |
| eClass | 8       | 27-04-90-02    |
| eClass | 7.1     | 27-04-90-02    |
| eClass | 6       | 27-04-90-02    |
| ETIM   | 10      | EC002540       |
| ETIM   | 9       | EC002540       |
| ETIM   | 8       | EC002540       |
| ETIM   | 7       | EC002540       |
| IDEA   | 4       | 4130           |
| UNSPSC | 15      | 39-12-10-04    |

Approvals Certificates

General Product Approval



[Manufacturer Declaration](#)[Declaration of Conformity](#)

General Product ApprovalMaritime applicationother

[Miscellaneous](#)[Miscellaneous](#)[BIS CRS](#)[Miscellaneous](#)

Environment



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